

Economics 880 – PSet 2

Aaron Banks & Arthur Rodrigues

1 Part 1

We solve for the competitive equilibrium in the setting of Hugget (1993) with enforceable insurance markets and no initial asset holdings. There are a unit measure of identical agents with preferences $\mathbb{E}_0[\sum_t \beta^t U(c_t)]$ where $U : \mathbb{R}_+ \rightarrow \mathbb{R}$ is strictly increasing, strictly concave, differentiable and bounded on its domain. As usual, $\beta \in (0, 1)$. The support of income shocks is $S = \{u, e\}$ where income is $y : S \rightarrow \mathbb{R}$ with $y(e) = 1$ and $y(u) = b < 1$. Shocks are iid across agents and follow a Markov process:

$$\pi(s' | s) = \Pr(s_{t+1} = s' | s_t = s).$$

Each period, agents' income is realized, and they choose asset holdings $a_{t+1} \geq \underline{a}$ given a price $q_t \in [0, 1]$ with consumption c_t determined by the residual income from asset holding decisions and where $\underline{a} \leq 0$.

We solve for the competitive equilibrium by first solving a planner's problem that weights each agent equally, and then decentralizing the planner's allocation via a bond price. While agents are subject to income risk, in aggregate the Law of Large Numbers implies no aggregate uncertainty. Thus, the planner's problem is given by

$$\max_{\{(c_t^e, c_t^u)\}} \sum_{t=0}^{\infty} \beta^t [\pi^e U(c_t^e) + \pi^u U(c_t^u)] \quad \text{s.t.} \quad \pi^e y(e) + \pi^u y(u) \leq \pi^e c_t^e + \pi^u c_t^u$$

where π^s is the fraction of the population that receives income shock s each period. With multiplier λ_t on the period t resource feasibility constraint, the first order optimality conditions of the associated Lagrangian are

$$\begin{aligned} \beta^t U'(c_t^s) - \lambda_t &= 0 \quad s = e, u \\ \lambda_t [\pi^e (c_t^e - y(e)) + \pi^u (c_t^u - y(u))] &= 0 \end{aligned}$$

Strict concavity of the utility function implies that $c_t^e = c_t^u = \pi^e y(e) + \pi^u y(u)$ for all t . In other words, there is perfect consumption smoothing between those agents suffering different income shocks.

The household problem is

$$\begin{aligned} \max_{\{(c_t, a_t)\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t U(c_t(s_t)) \quad \text{s.t.} \quad c_t + \sum_{s \in S} Q(s | s_t) a_{t+1}(s, s_t) &\leq y(s_t) + a_t(s_t, s_{t-1}) \\ a_0 &= 0 \end{aligned}$$

The consumption Euler equation for this problem is

$$Q(s' | s)U'(c_t(s)) = \beta\pi(s' | s)U'(c_{t+1}(s')).$$

To match the planner's allocation, the marginal utilities of consumption in adjacent periods must be equal as consumption is smoothed across time and states. Hence,

$$Q(s' | s) = \beta\pi(s' | s),$$

implying that the risk free rate is

$$q \equiv \sum_{s' \in S} Q(s' | s) = \beta.$$

2 Part 2

3 Part 3